Fleets of Certified AI-Load UPS Units for Oscillation Damping and Frequency Support on the Kundur Two-Area System

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Abstract: **A previous paper certified the grid-side behavior of one battery-backed UPS at an AI training campus: its grid draw changes no faster than a ramp limit, and its battery stays inside a computed energy window. Here several such units sit on one transmission network and, in addition to hiding the load swing, they are asked to damp inter-area oscillations and to support frequency after a generator trip, using PMU-style frequency measurements. I report what is proved and what is only simulated. Proved: (i) the per-unit certificate survives an arbitrary grid-service command inside a headroom band, for headroom up to 15 MW per 100 MW campus with a 20 MW authority; (ii) a passivity argument that a fleet of proportional frequency-feedback units cannot destabilize an idealized lossless network; (iii) rounding-safe eigenvalue enclosures and a common conic-sector certificate for the real lossy, PMU-filtered model; (iv) verified nadir bounds for the saturated closed loop after each generator trip; and (v) a forced-oscillation bound from the ramp limit. On the Kundur two-area system with two sites of three campuses each, the passive fleet raises the verified inter-area damping ratio from 0.024 to 0.106 and lifts every verified N-1 nadir by 0.14 to 0.15 Hz, but that is not enough to avoid the first under-frequency load-shedding stage, and the verified bound is optimistic by 0.07 to 0.11 Hz against a nonlinear simulation. A tuned power-system-stabilizer-style controller matches the passive one on nadir and gains 0.006 in damping, with no proof of its saturated behavior. A neural controller trained by evolution strategies, with no proof, has an unstable small-signal linearization in 9 of 11 test scenarios. The IEEE 39-bus run did not finish and is reported as incomplete.**

Index Terms: uninterruptible power supply, large loads, inter-area oscillations, primary frequency response, passivity, verified computation.

I. Introduction

Synchronized AI training jobs swing the power of a campus between a floor and a peak in about a second, and the campuses that do it are large: ERCOT defines a large load as 75 MW and up, and a 200 MW training site appears in the NERC data [7], [9]. The first paper in this series [1] treated one campus. It certified a UPS controller so that, for every load profile in a stated band, the grid draw obeys a ramp limit R and the battery energy stays inside a computed window. The certificate is a robustly invariant polytope found by a sound interval branch-and-bound search.

A fleet of such units is more than a set of smoothers. The same converter and battery can also inject or absorb a small amount of real power on command, and a grid operator would like that power to damp inter-area modes and to arrest frequency after a generator trip. Two questions follow. Does the per-unit guarantee survive when the unit also takes grid-service commands? And what can be said about the whole fleet, as opposed to one unit, without assuming that the controller is well behaved?

This paper answers with a mix of proofs and simulations, and it keeps the two separate. The test system is the Kundur two-area network [2], with two sites at its load buses and three 100 MW campuses at each. The baselines are a passive proportional controller, which is the unit the passivity result covers; a power-system-stabilizer-style lead-lag controller tuned by the residue method [3], [4], which has no proof here; and a small neural controller trained by evolution strategies, which also has none. Parameters that have a public source are taken from it and cited; the others are marked as assumed. Where a claim failed, it is reported as a negative result.

Scope. The IEEE 39-bus system was part of the plan. Its Kaggle run ended in an error status while solving the robust LMI for the learned controller (the LMI for POD alone had taken about 65 minutes), before any 39-bus damping, contingency or forced-oscillation file was written. I report only the three things that were saved (Section VII) and draw no conclusion from them.

II. Related Work

Low-inertia frequency stability is surveyed in [11]. Passivity of the swing equations and of frequency-feedback controllers on lossless networks is used in the power-system literature to argue stability without network knowledge, for example for primary frequency regulation in [12]; Proposition 1 is an instance of that argument with a converter lag. Power-oscillation damping with a stabilizer-style lead-lag is standard [3], [4] and is used here as the unproved baseline. The large-load literature [7]–[9] concerns the load swing itself and utility interconnection limits; the extra step here is to ask the load's own storage for grid service while keeping a certificate that does not depend on what controller issues the service command. Certified sets for grid-forming converters are in [13].

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III. Model and Sourced Parameters

A. Network and machines

The model keeps the angle and real-power coupling only. Machines are classical (constant $|E'|$ behind x'_d), the network buses are algebraic nodes at fixed voltage magnitude, and loads are constant power. Around the power-flow point of pandapower [6], with d the machine angles and θ the bus angles,

$$\theta = Wd + Zp_b, \qquad p_e = Kd + B_p p_b, \tag{1}$$

where p_b is the net bus injection (the UPS enters here) and p_e the machine electrical power. Machine i obeys $\dot{d}_i = \omega_s w_i$ and $M_i \dot{w}_i = p_{m,i} - p_{e,i} - D_i w_i$. The governor is TGOV1 without limits. Machine inertia, x'_d and TGOV1 data come from the case files shipped with ANDES [5] (variant of the Kundur system, file `kundur_full.xlsx`), as recorded in the sourcing step of this project. Both ANDES cases list $D = 0$, so the damping $D = 2$ pu used here stands in for damper windings and load damping and is *assumed*; $D \in [1, 3]$ is swept. Setting the conductances to zero makes Z symmetric positive definite, which is the setting of Proposition 1. Every reported eigenvalue and simulation uses the lossy model.

B. The UPS site

Each site k has the chain: PMU-style frequency estimate y_k, controller q_k, converter lag $T_u \dot{v}_k = -v_k - q_k$, and grid injection $u_k = \mathrm{sat}(v_k, \pm\bar{u}_k)$. The PMU estimate is a first-order angle tracker, $T_f \dot{\xi} = \theta - \xi$, $y = (\theta - \xi)/(T_f \omega_s)$. A site contains three campuses, so $\bar{u}_k = 3 \times 10 = 30$ MW. The UPS draws $P_g - u$ from the grid, with P_g the paper-1 law, so a positive u gives power back.

Table I lists the values. The efficiency and ramp limit are the two sourced changes that matter for the per-unit certificate; both are applied in every number below.

TABLE I
Parameters: sourced versus assumed. Sources are traced in the project's sourcing note.

Quantity	Value	Status
Grid ramp limit R	20 MW/min per campus	sourced: Southern Company, via [9] Table 1 (paper 1 assumed 10)
One-way efficiency η	$\sqrt{0.86} = 0.927$	sourced: LFP AC-AC round trip 86%, [10] Table 4.4 (paper 1 assumed 0.95)
UFLS first stage	59.3 Hz	sourced: Ameren, via [9] Table 2; used as a 0.7 Hz margin
Machine H, x'_d, TGOV1	case data	sourced: ANDES [5]; network and dispatch from [6]
Campus swing	40 MW (3 per site)	paper 1 value; consistent with 35 to 70 MW per minute swings in [7]
Swing content	0.2 to 3 Hz	sourced: [8]; used to choose the forced-oscillation sweep
Machine damping D	2 pu, swept 1 to 3	**assumed** (ANDES lists 0)
Converter lag T_u	50 ms	**assumed**; 100 ms tested
PMU tracker T_f	33 ms	**assumed**; 100 ms tested
Wide-area delay	50 ms	**assumed**; 0.2 and 0.5 s tested
Headroom $\bar{u}$	10 MW per campus	**assumed**; swept 0 to 15 MW
Campuses per site, site buses	3; pandapower buses 6, 7	**assumed**
Campus size, load band	100 MW, $[60, 100]$ MW	paper-1 setting (load floor ≈ 0.6 is consistent with [7], read off a plot)

IV. What Is Proved

A. Per-unit certificate with a service channel

Treat the grid-service command $u \in [-\bar{u}, \bar{u}]$ as a second adversarial, measurable input beside the load P_L. The paper-1 law for P_g is unchanged. A polytope that passes the paper-1 band condition for every (P_L, u) in the box is robustly invariant for *any* command with $|u| \le \bar{u}$, whatever produced it, including the neural controller. The check is the same sound interval branch-and-bound as in paper 1, now on four-dimensional boxes, and it also certifies the ramp limit and the battery window. A closed-form necessity test rules out $\bar{u} > c$, where c is the authority of the state-of-charge loop.

Table II gives the outcome with the sourced R and η. With $c = 20$ MW every tested headroom is certified, up to 15 MW. With $c = 10$ MW the certificate fails at 7.5 MW and above. At the fleet's nominal point ($c = 20$ MW, $\bar{u} = 10$ MW) the certified window is 3.48 MWh and the peak battery power is 70.3 MW, per campus. Headroom costs energy: the window grows from 1.01 MWh at $\bar{u} = 0$ to 3.48 MWh at 10 MW (Fig. 1).

TABLE II
Per-campus certificate versus service headroom (window in MWh, peak battery power in MW). Dash: not certified.

$\bar{u}$ (MW)	0	2.5	5	7.5	10	15
$c = 10$ window	1.006	1.551	2.183	–	–	–
$c = 10$ power	40.3	51.6	55.3	–	–	–
$c = 20$ window	1.015	1.271	2.353	2.888	3.483	4.638
$c = 20$ power	40.6	49.2	63.4	67.8	70.3	75.3

B. Passivity of the fleet

Proposition 1. *Take the lossless network ($G = 0$), machine damping $D_i \ge 0$, TGOV1 governors with $T_2(T_1 + T_3) \ge T_1 T_3$, ideal (unfiltered) bus-frequency measurement, and sites with $u_k = -K_k y_k$ passed through the lag $1/(1 + sT_u)$, $K_k \ge 0$, no saturation. Then the closed loop has no eigenvalue in the open right half plane, for every number and placement of sites and every $K_k \ge 0$.*

Proof sketch. Let V be the sum of the kinetic energy $\frac{1}{2}\sum M_i w_i^2$, the network potential $\frac{1}{2} d^\top K d$ (symmetric and positive semidefinite when $G = 0$), the governor storage, and $\frac{1}{2} u^\top Z u/\omega_s$. With $G = 0$, the power the sites inject into the network equals $u^\top \dot{\theta}/\omega_s$ and $\dot{\theta} = W\dot{d} + Z\dot{u}$ with $W = B_p^\top$, so the $Z\dot{u}$ term is the exact derivative of the last storage term and $\dot{V} = -\sum D_i w_i^2 - (\text{governor dissipation}) + u^\top y$, with y the bus

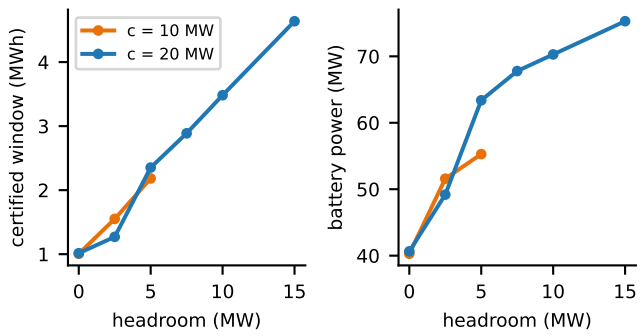


Fig. 1. Certified battery energy window and peak battery power per campus against service headroom. Missing points for $c = 10$ MW at 7.5 MW and above are designs the search could not certify.

frequency in per unit. The governor transfer $(1 + sT_2)/((1 + sT_1)(1 + sT_3))$ has real part $1 + \omega^2[T_2(T_1 + T_3) - T_1T_3]$ up to a positive factor, so the condition makes it positive real, hence dissipative. The unit map $y \mapsto -K/(1+sT_u)\, y$ is the negative of a positive-real map, so with $u = -\mathcal{K}(s)y$ the term $u^o py$ is nonpositive in the energy sense. A Lyapunov function that does not increase excludes open-right-half-plane eigenvalues. □

The governor condition holds for all four machines: $T_2(T_1 + T_3) = 15.73 \geq T_1T_3 = 3.43$ (TGOV1 data from ANDES). Two limits of this result matter. First, the real model has line resistance and a PMU filter. The loop $K/((1 + j\omega T_f)(1 + j\omega T_u))$ loses positive realness above $1/(2\pi\sqrt{T_fT_u}) = 3.92$ Hz, which is above the inter-area band, and the proposition does not cover either effect. Second, the proposition says stability, not damping: it guarantees no harm, not benefit. For the lossy, filtered model the stability claim rests on the verified enclosures below, not on this argument.

C. Verified small-signal facts

Eigenvalues of each closed-loop matrix A are enclosed with a Bauer–Fike bound [14]: from $AV = V\Lambda + R$ with the residual computed in floating point plus a rounding allowance, every eigenvalue lies within $\kappa_2(V)\|RV^{-1}\|_2$ of a diagonal entry of Λ. A damping ratio interval follows. For parameter boxes I use the conic-sector LMI of Chilali and Gahinet [15] with a single X over the vertex matrices, solved with Clarabel and then re-verified: the solution is accepted only if a floating-point Cholesky test passes with Rump's allowance for rounding [16]. The box is machine damping $D \in [1, 3]$ crossed with a fleet-gain scale in $[0.5, 1.5]$; A is affine in both, so four vertices cover the box.

D. Verified contingency bounds

After a generator trip the saturated fleet is piecewise affine. In each saturation region the flow is exact, $z(t) = \exp(\hat{A}_s t)z(0)$ with $z = [x; 1]$, and a step is accepted only if a Taylor enclosure shows that every UPS state stays inside the region on the step; otherwise the step is halved and the crossing is located by bisection. The resulting bound on each machine speed is an interval enclosure for the *model*, up to rounding in the matrix exponential (not enclosed). After 40 s a verified positive definite Lyapunov ellipsoid inside the final region bounds the tail. Nothing here covers the neglected voltage dynamics or the difference between linear and sine coupling; I measure that gap by simulation in Section VI-B.

E. Forced-oscillation bound

Paper 1 proves the grid draw of each campus is R-Lipschitz. A periodic R-Lipschitz signal at frequency f has a fundamental amplitude of at most $R/(\pi f)$. Multiplying by the linear-model gain from site injection to worst machine speed bounds the speed oscillation a swinging campus can cause, for the fundamental, with all sites in phase. The comparison case is an unsmoothed square-wave swing of the same size. This is a bound on the first harmonic only.

V. Controllers

Passive (proved). $q_k = Ky_k$ with one gain $K = 987$ pu power per pu speed at every site, chosen on an 80-point log grid from 1 to $2times10^4$ to reach the target damping $\zeta = 0.10$ in the 0.1 to 1 Hz band.

PSS-style POD (no saturated proof). $q_k = K_{\text{pod}} \frac{sT_w}{1+sT_w} \Phi(s)\, y_k$ with washout $T_w = 10$ s and two lead-lag stages, phase set at the inter-area mode (0.745 Hz) by the residue method and gain scaled (factor 0.137) to reach $\zeta = 0.10$ [3], [4]. Its closed loop is verified linearly like the others, but the linear conic certificate does not extend to its saturated, contingency behavior beyond what the piecewise simulation in Section VI-B verifies, and it is outside Proposition 1.

Learned (no proof). A one-hidden-layer tanh network (32 units, no biases so $q(0) = 0$) maps the local frequencies and four machine speeds delayed by 50 ms to the q_k, trained by antithetic evolution strategies (150 iterations, 24 pairs, Adam, 8 random step disturbances per iteration, 10 s horizon, nominal operating point only). The objective is inertia-weighted speed deviation from the center of inertia plus a penalty on u^2. There is no certificate and no use of the perturbed test scenarios in training.

All three feed the same converter lag and the same saturation, so the per-unit certificate of Section IV-A holds for all of them. That is the one guarantee that does not depend on the controller.

VI. Results on the Kundur System

A. Oscillation damping

Table III and Fig. 2 give the damping ratio of the least-damped mode in the 0.1 to 1 Hz band for eleven scenarios (load $\pm 10\%$, $D \in \{1, 3\}$, one line out, $T_f = 0.1$ s, $T_u = 0.1$ s, site 0 offline, and wide-area delay 0.2 and 0.5 s, which affect only the learned controller). For the three linear controllers the band mode is isolated and all damping intervals are tight (widths under 10^{-4} for passive, about 0.01 for POD).

Both stabilizing controllers raise the band damping ratio from 0.024 to about 0.11, and POD is ahead of passive in most

TABLE III
DAMPING RATIO OF THE LEAST-DAMPED MODE IN THE 0.1 TO 1 HZ BAND (VERIFIED INTERVALS), AND THE WEAKEST OSCILLATORY MODE OF ANY FREQUENCY (NONE / PASSIVE / POD / LEARNED). "PASSIVE IDEAL" KEEPS THE LOSSY NETWORK BUT USES THE EXACT BUS FREQUENCY INSTEAD OF THE PMU ESTIMATE. DASHES: NOT APPLICABLE (THE DELAY ONLY ENTERS THE LEARNED CONTROLLER) OR NO BAND MODE.

Scenario	none	passive	POD	passive ideal	weakest mode, all frequencies
nominal	0.024	0.106	0.112	0.119	0.012 / 0.019 / 0.020 / -0.050
load 0.9	0.024	0.104	0.109	0.116	0.012 / 0.019 / 0.020 / -0.043
load 1.1	0.024	0.109	0.116	0.123	0.012 / 0.020 / 0.020 / -0.059
D 1	0.015	0.098	0.103	0.111	0.007 / 0.014 / 0.015 / -0.162
D 3	0.032	0.115	0.121	0.128	0.017 / 0.025 / 0.025 / 0.044
line out 4	0.028	0.167	0.181	0.190	0.012 / 0.020 / 0.021 / -0.055
pmu Tf 0.1	0.024	0.079	0.093	–	0.012 / 0.016 / 0.017 / -0.045
conv Tu 0.1	0.024	0.086	0.098	0.098	0.012 / 0.016 / 0.018 / -0.041
site 0 offline	0.024	0.091	0.067	0.093	0.012 / 0.013 / 0.013 / 0.013
delay 0.2	–	–	–	–	– / – / – / -0.081
delay 0.5	–	–	–	–	– / – / – / -0.229

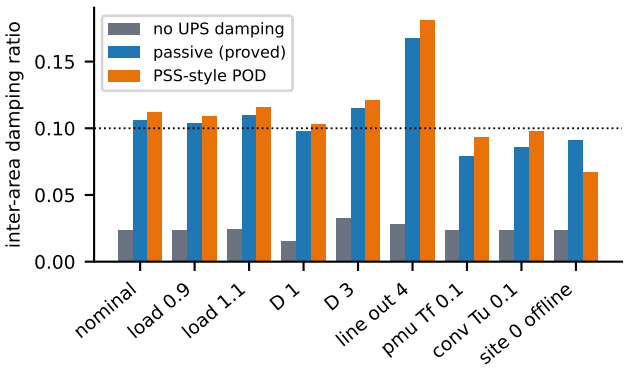


Fig. 2. Verified inter-area damping ratio across the nine linear-model scenarios. The dotted line is the 0.10 target. The learned controller is omitted: its linearization is unstable in 9 of 11 scenarios (Table III).

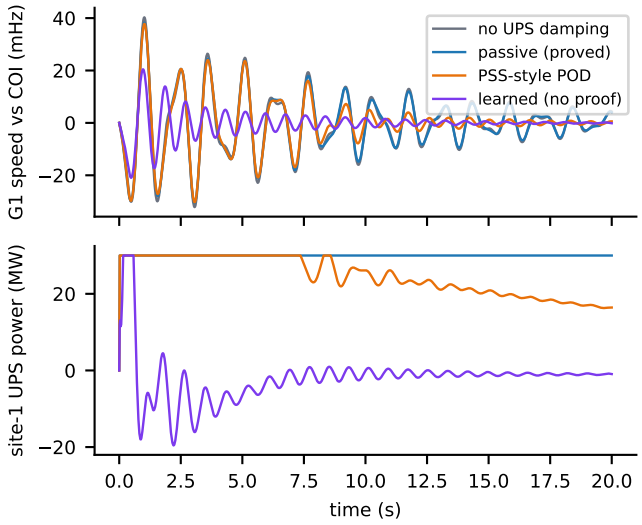


Fig. 3. Response to a 2 MW step at site 1 (linear model). Top: speed of G1 relative to the center of inertia. Bottom: site-1 UPS power. The learned controller's nominal-case response looks best here but it has no stability proof.

scenarios, by 0.006 at nominal and 0.014 under a slow PMU. POD falls behind with site 0 offline (0.067 versus 0.091), because the single remaining site cannot supply the gain the residue method assumed. The passive controller, which needs no tuning to the mode, is within 0.015 of POD in every scenario except site 0 offline.

The weakest mode did not improve much. The smallest damping ratio of any mode rises only from 0.012 to 0.019, so the benefit is in the band, not everywhere. The mode that sets the weakest ratio was not identified in this run. A 20 s simulation of a 2 MW step at site 1 shows the same thing: the inertia-weighted integral of squared speed deviation from the center of inertia falls by only 4.5% for passive and 20.9% for POD, with peak deviation 0.040 Hz (no UPS), 0.038 (passive), 0.038 (POD) (Fig. 3). The learned controller shows an 83% reduction in this one test, despite its unstable linearization (below).

Robust certificate. Over the (D, gain) box the common-X certificate gives a damping ratio of at least 0.0061 with no UPS (vertex minimum 0.0068), and 0.0098 for passive and 0.0098 for POD (vertex minimum 0.0109 for both). These numbers are about the weakest mode, so they show the same modest gain. The learned controller fails: its vertex minimum is -0.78, and no certificate exists.

Learned controller, negative result. Its small-signal linearization at the origin has a mode with negative damping ratio (-0.050) in nominal, load, $D = 1$, line-out, slow-PMU, slow-converter and both delay scenarios. Only $D = 3$ and site 0 offline have positive ratios (0.044 and 0.013 by weakest mode), and even there the eigenvalue enclosure is too loose to verify stability. The ES objective was still noisy and not converged: the mean cost fell from 4.20 (first ten iterations) to 3.02 (last thirty), with a best single value of 0.85, and I use the last iterate. A better-trained network might do better, so this is a result about this training run, not a bound on learned controllers. It does support the narrower point that nothing in the training objective forces small-signal stability.

B. N-1 frequency support

For each of the four machines I remove the unit at its dispatched output (700 to 727 MW) and compute the verified

TABLE IV
N-1 NADIR (HZ, DEVIATION FROM 60 HZ). VERIFIED BOUNDS FOR NONE, PASSIVE, POD; SIMULATED ONLY FOR LEARNED. "NL" IS A NONLINEAR-MODEL SIMULATION WITH SINE COUPLING, FOR NONE AND PASSIVE.

Trip	none	passive	POD	learned	NL none / pass.
G1 (727 MW)	−1.729	−1.585	−1.585	−1.656	−1.833 / −1.685
G2 (700)	−1.709	−1.565	−1.565	−1.708	−1.782 / −1.634
G3 (700)	−1.797	−1.651	−1.651	−1.803	−1.883 / −1.732
G4 (700)	−1.752	−1.605	−1.605	−1.727	−1.864 / −1.714

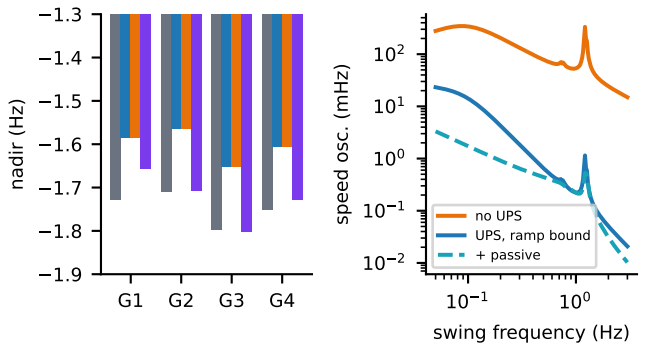


Fig. 4. Left: N-1 nadir by tripped machine (verified bounds; simulated for the learned controller). Right: forced-oscillation speed amplitude against swing frequency for 120 MW per-site square-wave swing (no UPS) and the ramp-limit bounds.

nadir bound over 40 s (Table IV, Fig. 4).

- The fleet lifts the verified nadir by 0.144 to 0.147 Hz, about 8%. In every trip the first UFLS stage (0.7 Hz) is passed, by 0.87 to 0.95 Hz with the fleet and by 1.01 to 1.10 Hz without it. **The fleet is too small to avoid load shedding in this model.** Each site saturates at 30 MW and stays there for the whole 40 s: the energy delivered by the passive fleet is 0.333 MWh per site, i.e., 0.111 MWh per campus, well inside the certified 3.48 MWh window.
- POD and passive give the same nadir to three decimals, because both are saturated while the frequency is low. The controller shape matters for damping, not for the nadir.
- The initial RoCoF bound is unchanged by the fleet (2.67 Hz/s for G1). The UPS lag is too slow to act on it.
- The learned controller's simulated nadir is worse than the verified passive bound in every trip, and equal to the no-UPS case for G2 and G3 within 0.01 Hz.
- The linear-model bound is optimistic. The nonlinear simulation is lower than the verified bound by 0.07 to 0.11 Hz in both columns. The fleet's *gain* is the same in the nonlinear model (0.148 to 0.151 Hz), which is the part I trust more than the absolute level.
- Imposing the sourced ramp limit on the UPS draw costs about 8 to 9 mHz of nadir (G1: −1.593 against −1.585 Hz).

C. Forced oscillations from the load itself

Three 40 MW campuses per site swing in phase, so 120 MW per site. Without a UPS the worst machine-speed amplitude over the sweep is 346 mHz at 0.087 Hz. With the R-Lipschitz bound ($R = 1$ MW/s per site) the bound is 16.6 mHz, a factor 20.9; with passive damping on top it is 1.97 mHz. At 0.25 Hz (the frequency of the 2019 Florida event in [7]) the three numbers are 162.5, 2.72 and 0.75 mHz; at 1.0 Hz (the 2023 Midwest event in [7]) they are 53.8, 0.223 and 0.219 mHz. These are bounds on the fundamental of the linear model. The comparison is to a square wave, so it measures the value of the ramp limit, not of the fleet's grid-service role, and the passive column helps mainly below 0.5 Hz, near the inter-area modes.

VII. IEEE 39-BUS: INCOMPLETE

The full run produced the 39-bus controllers and then ended with an error status in the robust-LMI stage for the learned controller, so there is no 39-bus damping, contingency, or forced-oscillation result file. Three saved values: with five sites (buses 3, 7, 15, 19, 38 in pandapower indexing) the open-loop band mode is 0.995 Hz with $\zeta = 0.1415$, already above the 0.10 target, so the passive-gain search ended on its lowest grid value ($K = 1$) and POD at its lowest scale (0.001, $\zeta = 0.1422$). The common-X certificate gives 0.0682 for none, passive and POD alike. In short, this 39-bus operating point has no band-damping problem for a fleet to solve, and nothing about the fleet on that system has been shown. The 39-bus nadir, the nonlinear gap, and the five-site gain scaling are open.

VIII. DISCUSSION AND LIMITS

What is proved. (a) The per-unit certificate with a service channel, with the sourced R and η. (b) A passivity statement for an idealized network. (c) Rounding-safe eigenvalue and conic-sector certificates for the lossy PMU model, over the stated box. (d) Verified nadir bounds for the model after each trip. (e) The forced-oscillation bound for the fundamental. **What is simulated only:** the learned controller, the nonlinear nadir, the time series, and everything on a different network.

Weaknesses.

1) The proofs are about the linear (or piecewise-affine) model. The nonlinear simulation lowers the nadir by 0.07 to 0.11 Hz, so the verified bound is not a bound on the real system.
2) Proposition 1 needs a lossless network and ideal measurement. For the real model I verify stability numerically, which does not extend to other networks.
3) The fleet is small. Sixty megawatts against a 700 MW trip cannot avoid load shedding; the improvement is 8%. The energy window is not binding. A sizing study would need real headroom data, which I could not find.
4) The weakest mode barely improves (0.012 to 0.019), and which mode that is was not identified.
5) The learned baseline is a short, single-seed training run and may be a weak baseline. The comparison shows only that an unconstrained objective produced a controller without stability.

6) Several values are assumed (Table I): machine damping, converter lag, PMU filter time, wide-area delay, headroom and the siting. Two have been varied (delay, T_f, T_u, D); headroom and siting have not.
7) Only one test system finished. The 39-bus result is missing, so any claim of scale beyond four machines and two sites is unsupported.

IX. Conclusion

On the Kundur two-area system, a fleet of certified AI-load UPS units with a simple proportional frequency feedback damps the inter-area mode (0.024 to 0.106, verified), bounds forced oscillations from their own load swings through the ramp limit, and adds 0.14 to 0.15 Hz to every N-1 nadir, while the per-unit battery certificate still holds for any command in the headroom band. The nadir gain is not enough to avoid load shedding, the verified level is up to 0.11 Hz optimistic against a nonlinear simulation, and a PSS-style controller buys damping but no frequency support at the cost of a proof. An unconstrained learned controller was unstable in small signal in 9 of 11 scenarios. The 39-bus extension is not done.

Reproducibility

Code and result files: `research/p2-fleet` in the project repository (`run_p2.py`, `results/`), run on Kaggle; the numbers here are read from `results/kundur.json`, `unit_cert.json` and `controllers_*.json`.

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